

Impact of Nonfiber Carbohydrate on Intake, Digestion, and Milk Production by Dairy Cows

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ABSTRACT

Four concentrations of dietary nonfiber carbohydrate (42, 36, 30, and 24%) were evaluated using eight multiparous Holstein cows (662 kg of BW; 63 DIM) in replicated 4 × 4 Latin squares with 28-d periods. Shelled corn and soybean meal were partially replaced with wheat middlings, dried brewers grains, and soy hulls to formulate 36, 30, and 24% nonfiber carbohydrate diets. Decreasing dietary nonfiber carbohydrate decreased DMI, did not affect actual or fat-corrected milk production, increased milk fat percentage, and decreased milk protein percentage and production. Apparent total tract DM and OM digestibilities were highest for 36% and lowest for 24% nonfiber carbohydrate diets. Digestibility of NDF was lowest for the 42% nonfiber carbohydrate diet. Ruminal pH and acetate:propionate increased, and total VFA concentrations decreased, as dietary nonfiber carbohydrate decreased. Ruminal degradation of alfalfa hay DM and NDF were higher for low nonfiber carbohydrate diets. Significant depression of DMI (9 to 1.8 kg/d) coincided with dietary nonfiber carbohydrate concentration at or below 30%. Results indicate that diets for cows producing 40 kg/d should contain more than 30% nonfiber carbohydrate and suggest little benefit of 42 over 36% nonfiber carbohydrate diets.

(Key words: nonfiber carbohydrate, intake, digestion, lactation)

Abbreviation key: NFC = nonfiber carbohydrate.

INTRODUCTION

Carbohydrate not recovered in NDF, referred to as nonfiber carbohydrate (NFC), includes starch, sugars, pectins, and β -glucans and can be estimated by subtracting CP, NDF, and ether extract from OM with a correction for CP bound to NDF (32). Optimal dietary NFC is not clearly defined.

Nocek and Russell (20) suggested that 40% dietary NFC was optimal in diets for lactating cows from an evaluation of diets based on alfalfa silage, corn silage, and 50:50 alfalfa: corn silage; dietary NFC ranged from 30 to 45%. Hoover and Miller (14) regressed data adapted from Nocek and Russell (20) and demonstrated that, when dietary NFC was greater than 45 to 50% or less than 25 to 30%, milk production was decreased. However, only small differences in milk production occurred as dietary NFC was varied from 40 to 35%. Sievert and Shaver (25, 26) reported no significant reductions in either DMI or milk production in response to reduction of dietary NFC from 42 to 35% with partial replacement of corn with fibrous carbohydrate sources. Other researchers (22, 29, 30) have reported no significant differences in DMI or milk production when percentages of dietary NFC were altered. Alteration of dietary NFC influenced ruminal fermentation patterns, total tract digestion of fiber, and milk fat percentage (25, 26, 30).

Partial replacement of corn with high fiber by-product feeds is a common practice on commercial dairies and can result in moderate to low NFC diets because high fiber by-products tend to be lower in NFC than corn (14). Data are limited on responses of lactating cows to moderate or low dietary NFC. The objectives of this study were to evaluate the effects of moderate to low dietary NFC on intake, digestion, and milk production in dairy cows.

Received November 10, 1993.

Accepted January 26, 1994.

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MATERIALS AND METHODS

Eight ruminally cannulated, multiparous Holstein cows averaging 662 kg of BW and 63 DIM at trial initiation were in two 4 × 4 Latin squares with 28-d periods. Squares were not run concurrently because of a lack of available cows; the second square was initiated 8 wk after the first. The first 14 d of each experimental period were for adaptation to diets; sampling was during d 15 to 28 of each period. Treatments were four NFC: 42, 36, 30, and 24% of dietary DM. Dried brewers grains, soy hulls, and wheat middlings partially replaced shelled corn and soybean meal to formulate diets containing 36, 30, and 24% NFC. Diets (Table 1) were formulated to contain 18.5% CP; 35% of CP as ruminally undegradable protein was calculated from NRC (19) and 21% NDF from forage (alfalfa silage). Diets met or exceeded NRC (19) mineral and vitamin requirements. Diets were fed as TMR twice daily at 0900 and 1800 h. Corn gluten meal and dried brewers grains were used to balance ruminally undegradable protein percentages.

Feed offered and orts were recorded daily. The amount of feed offered was adjusted daily to ensure 10% orts. Alfalfa silage was sampled weekly, and moisture content was determined by toluene distillation (9) for adjustment of diet formulations. Concentrate mixtures and alfalfa silage were sampled for nutrient analysis on d 15, 16, 22, and 23 of each period. Orts were sampled on d 24 to 26 of each period. Alfalfa silage, concentrate, and ort samples were dried for 48 h in a 60°C forced-air oven and ground to pass a 1-mm Wiley mill screen (Arthur H. Thomas, Philadelphia, PA). Alfalfa silage and concentrate samples were composited by period; orts were composited by cow within period. Feed and ort composites were analyzed for DM, OM, CP, ether extract (2), and for ADF and NDF (17). Estimation of NFC was by calculation (32), and measurement of starch and free glucose was by endoamylase and exoglucosidase incubation prior to using a glucose oxidase assay (13).

Cows were milked twice daily at 0600 and 1800 h, and production was recorded at each

TABLE 1. Ingredient and nutrient composition of experimental diets.

	NFC ¹			
	42%	36%	30%	24%
Ingredient, % of DM				
Alfalfa silage	47.5	47.7	47.7	47.5
Shelled corn	36.3	29.6	21.2	13.5
Soybean meal	9.5	5.8	.9	...
Corn gluten meal	1.4	1.7	2.6	...
Brewer's dried grains	3.4	6.9	9.4	19.6
Soyhulls	...	...	6.6	8.7
Wheat middlings	...	6.5	9.8	9.1
Dicalcium phosphate	.8	.7	.7	.4
Calcium carbonate	.2	.2	.2	.3
Trace-mineralized salt ²	.5	.5	.5	.5
Vitamin premix ³	.2	.2	.2	.2
Magnesium oxide	.2	.2	.2	.2
Nutrient, % of DM				
OM	91.6	91.4	91.3	91.0
CP	19.1	19.3	19.8	19.5
NDF	28.2	32.9	37.4	42.9
ADF	17.8	23.1	26.0	28.4
NDF from forage	20.5	20.5	20.5	20.5
Ether extract	3.6	3.8	3.7	3.8
NFC	40.7	35.4	30.4	24.8
Starch and free glucose	32.9	28.5	24.0	17.6

¹NFC = Nonfiber carbohydrate; NFC = OM - (CP + NDF + ether extract - CP bound to NDF).

²Contained .55% Mn, .55% Zn, .35% Fe, .14% Cu, .008% I, .006% Se, and .002% Co.

³Contained 2665 IU of vitamin A/g, 900 IU of vitamin D/g, and 3.52 IU of vitamin E/g.

milking. Milk weights recorded during d 14 to 28 of each period were used for data analysis. Fat and protein contents of milk sampled on week days of the last 2 wk of each period at both the a.m. and p.m. milkings were determined by infrared analysis (Wisconsin DHI Laboratory, Appleton). Mean daily milk composition was an average of a.m. and p.m. samples using the proportion of daily production at each milking as a weighting factor. Cows were weighed at 0900 h on 3 consecutive d at the start of the trial and on d 26 to 28 of each period.

Lanthanum was used as an external marker for apparent total tract digestibility measurements. A La solution (12) was sprayed onto soybean meal, and then this mixture was added to gelatin capsules (25) for oral administration twice daily at 0800 and 2000 h on d 16 to 26 of each period to provide approximately 35 mg of La/kg of dietary DM. A solution containing 1 g of Yb (24) was sprayed onto 500 g of alfalfa silage DM for estimation of forage passage rate. Solutions containing 1 g of Sm (12) and 3 g of Cr-EDTA (4) sprayed onto 500 g of DM for each concentrate mixture were used to estimate passage rates of concentrate and liquid, respectively. Markers were applied to feedstuffs by spraying rather than soaking because the soaking procedure might remove soluble feed DM, thereby altering its digestibility and passage kinetics. Also, differences in marker retention between spray and soaking techniques appear to be small (8). Marked alfalfa silage and concentrate mixtures were fed at 0900 h on d 23, immediately before the a.m. feeding. Fecal grab samples were collected before digesta marker dosing and 12, 18, 24, 36, 48, 60, and 72 h after dosing. Fecal samples were dried in a forced-air oven at 60°C for 72 h and ground through a 1-mm Wiley mill screen. A fecal composite for digestibility determination was made for each cow within period and analyzed for DM, OM, CP, ADF, NDF, and starch as previously described. Concentrations of La (fecal composites), Yb, and Sm (individual time points), were determined after wet ashing using nitric acid digestion for analysis by direct current plasma emission spectroscopy (7). Apparent total tract nutrient digestibilities were calculated 1) from La concentration in diet and feces and 2) from nutrient intakes. Fractional passage rates of forage,

concentrate mixtures, and liquid were determined by regression of the natural log of the rare earth concentration from the declining portion of the fecal excretion curve versus time.

Ruminal fluid was sampled immediately prior to the a.m. feeding and at 2, 4, 6, and 8 h postfeeding on d 22 of each period. Samples were obtained via the ruminal cannula by passage of a hand-capped plastic jar (250 ml) through the fiber mat and collection of fluid from the ventral rumen. Samples were strained through two layers of cheesecloth for immediate pH determination (Beckman model 1009 pH meter, combination electrode 39848; Beckman Instruments, Fullerton, CA). A 50-ml aliquot was acidified with 1 ml of 50% of H_2SO_4 for VFA analysis. A second 50-ml aliquot was acidified with 1 ml of 50% TCA solution for ammonia analysis. Both VFA and ammonia samples were immediately frozen at -20°C and were kept frozen until analyzed as described by Sievert and Shaver (25, 26).

Early bloom alfalfa hay (20% CP; 35% NDF) and the four concentrate mixtures were evaluated *in situ*. Dacron bags (19 × 9.5 cm with an average pore size of 52 μm), prepared as described by Shaver et al. (24), were filled with 6-g (as fed) samples of test feeds ground to pass a 2-mm Wiley mill screen. Bags were prepared in duplicate for each feed at each time point. The type of concentrate mixture used in the dacron bag incubation for each cow corresponded to the type of concentrate being fed to that cow during that period. Bags containing alfalfa hay were also incubated within each cow. Bags were incubated ruminally for 2, 4, 6, 9, 12, 24, 48, and 72 h. Immediately prior to being placed in the rumen, bags were soaked in tepid tap water for 30 min. Bags were placed in a weighted mesh net secured to the ruminal cannula and were introduced into the rumen in reverse order starting on d 24 of each period. All bags were removed at 0900 h on d 27 of each period and were washed twice in cold tap water in a commercial washer (5). Zero-hour bags were prepared for each feed type during each period in triplicate and were also immersed in tepid tap water for 30 min before washing. Duplicate empty bags were incubated in each cow to estimate influx of DM into the bags. Bags and residue were then weighed after drying for 72 h in a 60°C forced-

air oven. Rate and extent of DM disappearance were estimated using the NLIN procedures of SAS (23) as described by Sievert and Shaver (25). The rapidly disappearing fraction (A) was estimated as the disappearance from the 0-h bag, the undegraded fraction (C) as the residue remaining at 72 h, and the slowly disappearing fraction (B) as $100 - (A + C)$. The NDF content of alfalfa hay residues remaining in 12-, 24-, and 72-h bags were measured using the procedure of Mertens (17). Disappearance of NDF from the bags was estimated as the percentage remaining at these time points.

Data were analyzed as a replicated 4×4 Latin square using the general linear models procedure of SAS (23); square, cow and period nested within square, and treatment were the effects in the model. All terms were tested using the residual mean square error. For ruminal measurements, a split-plot over time model was used. Cow and period nested within square by diet, hour, and hour by diet terms were added to the model. Effects of square, cow and period nested within square, and diet were tested using cow by period nested within square by diet as an error term, and hour and hour by diet interactions were tested using the residual mean square error. No data were collected for one of the cows on the 42% NFC diet during Period 1 because she was off feed. Significance of effects was detected using an *F* test of Type III partial sums of squares, which adjusts analyses for missing data. Orthogonal polynomial contrasts were used to test for linear, quadratic, and cubic relationships between treatment and response. Significance of effects was designated as $P < .05$ unless otherwise noted.

RESULTS AND DISCUSSION

Diets (Table 1) were formulated to contain 18.5% CP on a DM basis but averaged 19.4% CP and ranged from 19.1 to 19.8% CP because the CP percentages of alfalfa silage, wheat middlings, and soy hulls were higher than values obtained from analysis and from ingredient composition tables (19) prior to trial initiation. Alfalfa silage averaged 47.4% DM, 19.6% CP, 43.1% NDF, 24.3% NFC, and 5.2% starch and free glucose (the latter all on a DM basis). Dietary NFC percentages of 40.7,

35.4, 30.4, and 24.8 in DM were close to formulation target percentages. Differences between formulated and actual dietary NFC values were greatest for the high NFC diet (42.0 vs. 40.7% of DM), which may have been related to the large amount of shelled corn in this diet and the difficulty in extracting starch during NDF analysis (32), thus lowering the NFC estimate. Differences between NFC and starch and free glucose averaged 7.1% units across the four diets, which can be partially explained by the inclusion of pectin from alfalfa in the calculated NFC values, but not in the assayed starch and free glucose values. The NDF increased from 28.2 to 42.9% of DM as dietary NFC declined from 40.7 to 24.8% of DM, although NDF from forage was constant (20.5% of DM).

The DM, NDF, and NFC intakes (adjusted for orts) and BW means are presented in Table 2. The BW averaged 688 kg over the duration of the trial and was similar for all treatments. Decreases in dietary NFC had a negative linear effect on DMI (kilograms per day and as a percentage of BW), possibly related to the higher NDF content of the low NFC diets. MacGregor et al. (15) observed higher DMI as dietary starch was increased from 25 to 33% of DM. This change in dietary starch was of similar magnitude to that observed for our 30 versus 42% NFC diets. Declines in DMI for 36, 30, and 24% NFC diets relative to the 42% NFC diet were .4, .9, and 1.8 kg/d, respectively. The small difference in DMI between 42 and 36% NFC diets is in agreement with results of Sievert and Shaver (25, 26).

Source of fibrous carbohydrate (dried brewers grains, soy hulls, and wheat middlings) used to replace part of the corn in this trial may possibly have affected intake, particularly for the 24% NFC diet, which contained a large proportion (37.4%) of these feeds. However, results of previous trials with these feed ingredients do not support this premise. Polan et al. (21) observed higher DMI for diets containing 11.0, 17.4, and 24.2% dried brewers grains than for soybean meal controls with similar percentages of CP. The proportion of dried brewers grains in the 24% NFC diet was lower than in their high dried brewers grains diet (19.6 vs. 24.2%). Acedo et al. (1) observed no depression of DMI when wheat middlings (20, 40, or 60% of the concentrate) replaced corn

TABLE 2. Effect of nonfiber carbohydrate (NFC) percentage on nutrient intakes, BW, milk production, and composition.

Item	NFC				SE
	42%	36%	30%	24%	
DMI					
kg/d	27.6	27.2	26.7	25.8	.2 ^L
% of BW	4.01	3.94	3.89	3.75	.03 ^L
NDF Intake					
kg/d	7.8	9.0	10.0	11.1	.1 ^L
% of BW	1.14	1.30	1.45	1.62	.02 ^L
NFC Intake					
kg/d	11.2	9.6	8.1	6.4	.1 ^L
% of BW	1.62	1.39	1.19	.93	.02 ^L
BW, kg	688	690	687	688	2
Production, kg/d					
Milk	40.2	39.7	39.7	39.5	.4
4% FCM	35.5	35.6	35.9	35.6	.4
Milk fat	1.29	1.32	1.33	1.32	.02
Milk protein	1.23	1.20	1.20	1.16	.01 ^L
Composition, %					
Milk fat	3.24	3.30	3.33	3.35	.03 ^L
Milk protein	3.07	3.02	3.02	2.95	.03 ^L

^LL = Linear effect ($P < .05$).

and sorghum in a pelleted concentrate. However, Bernard and McNeill (3) reported a significant decrease in DMI when wheat middlings were fed at 22% of DM in a TMR. The proportion of wheat middlings in the 30 and 24% NFC diets were lower than those fed in their trial (9.8 and 9.1 vs. 22%). Partial replacement of corn with soy hulls also has resulted in similar intakes (18, 25).

Decreased dietary NFC had a positive linear effect on NDF intake (kilograms per day and as a percentage of BW) because of the higher dietary NDF content. When expressed as a percentage of BW, intake of NDF averaged 1.14, 1.30, 1.45, and 1.62% for dietary NDF percentages of 28.2, 32.9, 37.4, and 42.9%, respectively. Reduction in DMI for 42 versus 36% NFC diets was small (.4 kg/d) relative to the increase of NDF intake (7.8 vs. 9.0 kg/d). Mertens (16) reported that daily NDF intake was 1.2% (SE = .1) of BW in diets that produced maximum 4% FCM production and suggested that this NDF intake constant could be used in models to predict DMI of diets containing mixtures based on ground corn and soybean meal. Intake of NDF was within this range for 36 and 42% NFC diets fed in our trial (1.30 vs 1.14% of BW) and the trial of

Sievert and Shaver [(25); 1.32 vs. 1.09% of BW], and DMI differences were small in both trials. Sievert and Shaver (26) observed significantly lower DMI (.9 kg/d) for 35 than for 42% NFC diets; NDF intakes were 1.49 and 1.22% of BW for low and high NFC diets, respectively. Depression of DMI in our trial relative to the 42% NFC diet at NDF intakes of 1.45 and 1.62% of BW for the 30 and 24% NFC diets, respectively, were .9 and 1.8 kg/d. Lactating cows can apparently tolerate fairly high dietary NDF without a major reduction of DMI when high fiber by-products are included in diets containing 21% NDF from alfalfa silage, although, at NDF intakes at or above 1.45% of BW, significant depression of DMI occurred. Decreasing dietary NFC had a negative linear effect on NFC intake (kilograms per day and as a percentage of BW). Intake of NDF plus NFC provides an estimate of total carbohydrate intake and averaged 19.0, 18.6, 18.1, and 17.5 kg/d and 2.76, 2.70, 2.63, and 2.54% of BW for 42, 36, 30, and 24% NFC diets, respectively. These values followed a trend that was similar to that for DMI, and the reduction was of similar magnitude, suggesting that intake of NDF plus NFC may be a better predictor of DMI than either dietary NFC or NDF alone for diets containing 21% NDF from alfalfa silage.

TABLE 3. Effect of nonfiber carbohydrate (NFC) percentage on apparent total tract nutrient digestion.

Item	NFC				SE
	42%	36%	30%	24%	
DM, %	72.3	73.2	71.7	68.3	.7 ^{LL,Q} ¹
OM, %	72.8	73.9	72.6	68.9	.7 ^{LL,Q}
CP, %	76.0	78.2	78.9	77.6	.6 ^{L,Q}
NDF, %	52.9	57.7	58.9	57.3	1.4 ^{L,Q}
ADF, %	58.0	58.7	58.7	56.6	1.2
Starch, %	95.0	96.8	98.2	98.7	.5 ^{LL}

¹L = Linear effect ($P < .05$), LL = linear effect ($P < .01$), and Q = quadratic effect ($P < .01$).

Milk production and composition data are summarized in Table 2. Milk and 4% FCM production were not affected by dietary NFC percentage. Other researchers (22, 25, 26, 29, 30) have reported no significant differences in milk production when percentages of dietary NFC were altered. This result is in contrast to that of Nocek and Russell (20), who observed highest milk production for 40% NFC diets from an evaluation with dietary NFC ranging from 30 to 45%. Lack of a milk production decline in our trial for 30 and 24% NFC diets is surprising when the magnitude of the DMI depression (.9 to 1.8 kg/d) in response to these treatments is considered. Further, intake of digestible OM was .6 and 2.0 kg/d lower for the 30 and 24% NFC diets. Continuous lactation trials may have revealed adverse effects of low NFC diets on changes in BW and body condition score that were related to the depression of energy intake. The BW change averaged +.23 kg/d and was similar for all treatments, but potential differences in ruminal fill between diets related to changes in NDF intake, and relatively short experimental periods (28 d) preclude evaluation of dietary NFC effects on energy balance.

Decreasing dietary NFC had a positive linear effect on milk fat percentage. This relationship has been reported by others (22, 26, 30) as being partially due to changes in ruminal pH and VFA. Production (kilograms per day) of milk fat was not affected by treatment. Decreased dietary NFC had a negative linear effect on milk protein percentage and production, possibly because of decreased microbial growth and protein synthesis in response to reduced ruminally available carbohydrate (27, 28). Milk protein percentage has been reported

to decrease as grain in the diet decreased (31). Our results show that adverse effects of low NFC diets on milk protein percentage and production are also of concern when high fiber by-product feeds partially replace corn in diets with a fixed forage:grain ratio.

Apparent total tract nutrient digestibilities are summarized in Table 3. Dietary NFC had linear and quadratic effects on DM and OM digestibilities that were highest for 36% NFC and lowest for 24% NFC diets. Linear and quadratic effects of dietary NFC were observed for CP and NDF digestibilities, which were highest for 30% NFC and lowest for 42% NFC diets. Higher digestibility of CP for the 30% NFC diet was possibly due to the higher dietary CP content (19.8 vs. 19.1%). Higher digestibility of NDF for moderate to low NFC diets has been reported by others (25, 26, 30) and may be due to higher NDF digestibilities for high fiber by-products than for forage. Digestibility of ADF was not affected by treatment. Decreases in dietary NFC had a positive linear effect on digestibility of starch. Higher total tract digestibilities of NDF and starch for moderate to low NFC diets were possibly due to positive associative effects of fiber on ruminal fermentation.

Ruminal fermentation data are presented in Table 4. Means presented are the average of samples taken at 0, 2, 4, 6, and 8 h postfeeding because no significant NFC by hour interactions were detected. Ruminal pH increased linearly, and total VFA concentration decreased linearly, as dietary NFC decreased, but, as reported by others (25, 26), no differences occurred between 42 and 36% NFC diets. Ruminal pH was above 6.0, and acetate:propionate was above 3.0 for all diets. Acetate

TABLE 4. Effect of nonfiber carbohydrate (NFC) percentage on ruminal pH, ammonia, and VFA.¹

Item	NFC				SE
	42%	36%	30%	24%	
pH	6.15	6.15	6.20	6.23	.01 ^{L2}
NH ₃ , mg/dl	25.5	24.9	24.0	26.7	.6
Total VFA, mM	112.1	115.8	108.4	99.6	3.3 ^L
VFA, mol/100 mol					
Acetate (A)	60.9	61.4	61.7	62.6	.2 ^L
Propionate (P)	20.2	19.5	19.6	19.3	.2 ^L
Butyrate	11.9	12.0	11.9	11.2	.1 ^{L,Q}
Others ³	7.1	7.2	6.8	7.0	.2
A:P	3.1	3.2	3.2	3.3	.1 ^L

¹Mean of ruminal fluid samples taken at 0, 2, 4, 6, and 8 h postfeeding.²L = Linear effect ($P < .05$) and Q = quadratic effect ($P < .01$).³Isobutyrate, isovalerate, and valerate.

(moles/100 mol) increased linearly, and propionate (moles/100 mol) decreased linearly as dietary NFC decreased. Sutton et al. (30) reported similar changes in acetate and propionate as dietary NFC decreased. Linear and quadratic effects existed for dietary NFC on butyrate, but the changes were small (12.0 vs. 11.2 moles/100 mol) and probably not of much biological significance. Ruminal ammonia and branched-chain VFA concentrations were not affected by treatment.

In situ estimates of DM degradation of alfalfa hay and concentrate mixtures are presented in Table 5. Percentages of rapidly and slowly degraded DM and ruminal availability of DM were highest for the concentrate fed in the 42% NFC diet, probably because of the lower fiber and higher starch content of that concentrate mixture. Rate of degradation and ruminal availability of alfalfa hay DM were highest for the 24% NFC diet. Percentages of alfalfa hay NDF remaining in dacron

TABLE 5. Effect of nonfiber carbohydrate (NFC) percentage on in situ DM degradation constants of concentrate mixtures and alfalfa hay.

Item	NFC				SE
	42%	36%	30%	24%	
Concentrate DM					
Rapidly degraded (A), %	33.8	29.7	27.0	26.2	.4 ^{L,Q2}
Slowly degraded (B), %	63.6	59.6	60.5	58.2	.8 ^{L,C}
Undegraded, %	2.6	10.7	12.5	15.6	.7 ^{L,Q}
Lag time, h	.9	.7	.6	.8	.3
Rate of degradation, /h	.05	.07	.05	.06	.01
Ruminal availability, % ¹	60.8	57.9	52.3	52.0	1.0 ^{L,C}
Alfalfa hay DM					
Rapidly degraded, %	28.6	27.6	28.0	28.0	.2
Slowly degraded, %	44.7	46.0	44.5	45.1	.4 ^C
Undegraded, %	26.7	26.4	27.5	26.9	.3
Lag time, h	2.8	1.5	2.2	1.8	.3
Rate of degradation, /h	.07	.07	.08	.10	.01 ^L
Ruminal availability, % ¹	52.0	51.9	53.6	55.4	.7 ^L

¹Ruminal availability = $A + B[k_d/k_d + k_p]$, using measured $k_p = .073/h$ for concentrate mixtures and assuming $k_p = .06/h$ for alfalfa hay, where k_d = rate of degradation and k_p = rate of passage (33).²L = Linear effect ($P < .05$), Q = quadratic effect ($P < .05$), and C = cubic effect ($P < .05$).

bags (data not provided in tables) decreased linearly as dietary NFC decreased from 76.2 to 66.9% and from 58.4 to 53.6% after 12 and 24 h of incubation, respectively. Mean percentage of alfalfa hay NDF remaining after 72 h of incubation was 44.2% and was not affected by treatment. Increases in ruminal degradation of alfalfa hay DM and NDF for low NFC diets were probably related to the higher ruminal pH and acetate:propionate ratio for these treatments. Low ruminal pH inhibited cellulolytic activity of rumen microbes (11).

Relative fractional rates of ruminal liquid passage, as estimated by a single ruminal dose of Cr (data not provided in tables), averaged .093/h and were similar for all treatments. Some (6, 26), but not all (25), researchers have reported increased liquid turnover as dietary fiber increased. Measures of fractional digesta passage rates are relative rather than absolute because rare earth markers may migrate from the feedstuff originally marked, and corrections for marker movement are not feasible (8). Fractional passage rates of Yb applied to alfalfa silage and SM applied to concentrate mixtures (data not provided in tables) averaged .058 and .073/h, respectively, and were similar for all treatments. Sievert and Shaver (26) reported that fractional passage rates of forage and concentrate were not affected by percentage of dietary NFC. Erdman et al. (10) observed no differences in rates of passage among corn, soybean meal, and high fiber by-product feeds.

CONCLUSIONS

Lactating cows can apparently tolerate fairly high dietary NDF concentrations and intakes of NDF without a major reduction of DMI when high fiber by-products are included in diets based on alfalfa silage, although, at NDF intakes at or above 1.45% of BW, DMI was significantly depressed. This depression coincided with dietary NFC concentrations at or below 30%. Production of milk and 4% FCM was not affected by dietary NFC concentrations, but continuous lactation trials may have revealed adverse effects of 30 and 24% NFC diets on changes in BW and body condition score that were related to the depression of DMI. Changes in ruminal pH and fermentation end products, forage NDF degradation,

and apparent total tract nutrient digestibility can be achieved by lower dietary NFC concentrations, resulting in increases in milk fat percentage. However, decreased dietary NFC decreased milk protein percentage and production. Results indicate that diets for cows producing 40 kg/d should contain more than 30% NFC and suggest little benefit of 42 over 36% NFC diets.

ACKNOWLEDGMENTS

The authors thank Leland Danz and Jim Armbruster for the feeding and care of the cows, Sandra Bertics for technical support in the laboratory, Michael Seaman and Kari Jensen for assistance with the analysis of feedstuffs, and Tom Tabone for assistance with the data analysis.

REFERENCES

- 1 Acedo, C., L. J. Bush, and G. D. Adams. 1987. Responses of dairy cows to different amounts of wheat middlings in the concentrate mixture. *J. Dairy Sci.* 70:635.
- 2 Association of Official Analytical Chemists. 1990. *Official Methods of Analysis*. 15th ed. AOAC, Washington, DC.
- 3 Bernard, J. K., and W. W. McNeill. 1991. Effect of high fiber energy supplements on nutrient digestibility and milk production by lactating dairy cows. *J. Dairy Sci.* 74:991.
- 4 Binnerts, W. T., A. T. van't Klooster, and A. M. Frens. 1968. Soluble chromium indicator measured by atomic absorption in digestion experiments. *Vet. Rec.* 82:470.
- 5 Cherney, D.J.R., J. A. Patterson, and R. P. Lemenager. 1990. Influence of in situ bag rinsing technique on determination of dry matter disappearance. *J. Dairy Sci.* 73:391.
- 6 Collucci, P. E., L. E. Chase, and P. J. Van Soest. 1982. Feed intake, apparent diet digestibility, and rate of particulate passage in dairy cattle. *J. Dairy Sci.* 65:1445.
- 7 Combs, D. K., and L. D. Satter. 1992. Determinations of marker in digesta and feces by direct current plasma spectroscopy. *J. Dairy Sci.* 75:2176.
- 8 Combs, D. K., R. D. Shaver, and L. D. Satter. 1992. Retention of rare earths by hay particles following incubation in fresh or autoclaved rumen fluid. *J. Dairy Sci.* 75:132.
- 9 Dewar, W. A., and P. McDonald. 1961. Determination of dry matter in silage by distillation with toluene. *J. Sci. Food Agric.* 12:790.
- 10 Erdman, R. A., J. H. Vandersall, E. Russek-Cohen, and G. Switalski. 1987. Simultaneous measures of rates of ruminal digestion and passage of feeds for prediction of ruminal nitrogen and dry matter digestion in lactating dairy cows. *J. Anim. Sci.* 64:565.

- 11 Grant, R. J., and D. R. Mertens. 1992. Development of buffer systems for pH control and evaluation of pH effects on fiber digestion in vitro. *J. Dairy Sci.* 75: 1581.
- 12 Hartnell, G. F., and L. D. Satter. 1979. Determination of rumen fill, retention time and ruminal turnover rates of ingesta at different stages of lactation in dairy cows. *J. Anim. Sci.* 48:381.
- 13 Herrera-Saldana, R., R. Gomez-Alarcon, M. Torabi, and J. T. Huber. 1990. Influence of synchronizing protein and starch degradation in the rumen on nutrient utilization and microbial synthesis. *J. Dairy Sci.* 73:142.
- 14 Hoover, W. H., and T. K. Miller. 1991. Associative effects of alternative feeds. Page 129 in *Proc. Alternative Feeds for Dairy and Beef Cattle*. Natl. Invit. Symp. St. Louis, MO. Coop. Ext., Univ. Missouri, Columbia.
- 15 MacGregor, C. A., M. R. Stokes, W. H. Hoover, H. A. Leonard, L. L. Junkins, Jr., C. J. Sniffen, and R. W. Mailman. 1983. Effect of dietary concentration of total nonstructural carbohydrate on energy and nitrogen metabolism and milk production in dairy cows. *J. Dairy Sci.* 66:39.
- 16 Mertens, D. R. 1987. Predicting intake and digestibility using mathematical models of ruminal function. *J. Anim. Sci.* 64:1548.
- 17 Mertens, D. R. 1991. Neutral detergent fiber. Page A12 in *Proc. Natl. Forage Testing Assoc. Forage Analysis Workshop*, Milwaukee, WI. Coop. Ext., Univ. Wisconsin, Madison.
- 18 Nakamura, T., and F. G. Owen. 1989. High amounts of soyhulls for pelleted concentrate diets. *J. Dairy Sci.* 72:988.
- 19 National Research Council. 1989. *Nutrient Requirements of Dairy Cattle*. 6th rev. ed. Natl. Acad. Sci., Washington, DC.
- 20 Nocek, J. E., and J. B. Russell. 1988. Protein and energy as an integrated system. Relationship of ruminal protein and carbohydrate availability to microbial synthesis and milk production. *J. Dairy Sci.* 71:2070.
- 21 Polan, C. E., T. A. Herrington, W. A. Wark, and L. E. Armentano. 1985. Milk production response to diets supplemented with dried brewers grains, wet brewers grains, or soybean meal. *J. Dairy Sci.* 68:2016.
- 22 Robinson, P. H., S. Tamminga, and A. M. Van Vuuren. 1987. Influence of declining level of feed intake and varying the proportion of starch in the concentrate on milk production and whole tract digestibility in dairy cows. *Livest. Prod. Sci.* 17:19.
- 23 SAS® User's Guide: Statistics, Version 5 Edition. 1985. SAS Inst., Inc., Cary, NC.
- 24 Shaver, R. D., A. J. Nytes, L. D. Satter, and N. A. Jorgensen. 1986. Influence of level of feed intake and forage physical form on digestion and passage of prebloom alfalfa hay in dairy cows. *J. Dairy Sci.* 69: 1545.
- 25 Sievert, S. J., and R. D. Shaver. 1993. Carbohydrate and *Aspergillus oryzae* effects on intake, digestion, and milk production by dairy cows. *J. Dairy Sci.* 76: 245.
- 26 Sievert, S. J., and R. D. Shaver. 1993. Effect of nonfiber carbohydrate level and *Aspergillus oryzae* fermentation extract on intake, digestion, and milk production in lactating dairy cows. *J. Anim. Sci.* 71: 1032.
- 27 Spicer, L. A., C. B. Theurer, J. Sowe, and T. H. Noon. 1986. Ruminal and post-ruminal utilization of nitrogen and starch from sorghum grain-, corn-, and barley-based diets by beef steers. *J. Anim. Sci.* 62:521.
- 28 Stern, M. D., H. Hoover, C. J. Sniffen, B. A. Crooker, and P. H. Knowlton. 1978. Effects of nonstructural carbohydrate, urea and soluble protein levels on microbial protein synthesis in continuous culture of rumen contents. *J. Anim. Sci.* 47:944.
- 29 Stokes, S. R., W. H. Hoover, T. K. Miller, and R. Blauweikel. 1991. Ruminal digestion and microbial utilization of diets varying in type of carbohydrate and protein. *J. Dairy Sci.* 74:871.
- 30 Sutton, J. D., J. A. Bines, S. V. Morant, and D. J. Napper. 1987. A comparison of starchy and fibrous concentrates for milk production, energy utilization and hay intake by Friesian cows. *J. Agric. Sci. (Camb.)* 109:375.
- 31 Tessman, N. J., H. D. Radloff, J. Kleinmans, T. R. Dhiman, and L. D. Satter. 1991. Milk production response to dietary forage:grain ratio. *J. Dairy Sci.* 74: 2696.
- 32 Van Soest, P. J., J. B. Robertson, and B. A. Lewis. 1991. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. *J. Dairy Sci.* 74:3583.
- 33 National Research Council. 1985. *Ruminant Nitrogen Usage*. Natl. Acad. Sci., Washington, DC.

REFERENCE ADDED PROOF

- 33 National Research Council. 1985. *Ruminant Nitrogen Usage*. Natl. Acad. Sci., Washington, DC.